

Md Farhamdur Reza

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PROFILE

Ph.D. researcher (ECE, NC State) specializing in **adversarial robustness**, **transferable/generative attacks**, and **federated learning**. First-author at **ICLR** and **ICCV** on query-efficient score/decision-based and curvature-aware attacks; building reproducible evaluation pipelines for vision and multimodal models. Current work includes responsible **LLM/VLM jailbreak evaluation** using single corrupted-prompt exposures.

SKILLS

ML/Research: Adversarial ML, AI Safety & Robustness, Alignment (RLHF, DPO, ORPO, SFT), Jailbreak Attacks, Adversarial Training, Generative AI, Federated Learning.

Frameworks: PyTorch, Hugging Face (Transformers, PEFT, TRL, Diffusers), TensorFlow, LangChain, scikit-learn, OpenCV; RobustBench, AutoAttack, Foolbox.

Programming/Systems: Python, CUDA, C/C++, MATLAB; Linux, Git, LSF; LaTeX.

RESEARCH EXPERIENCE

Graduate Research Assistant, Department of ECE, NC State University, Raleigh, NC **Fall 2021 - Present**

- **GSBA-K (ICLR'25):** Proposed a Top-K Geometric Score-Based Black-Box Attack that efficiently manipulates multiple top predictions simultaneously using novel gradient estimators.
- **CGBA (ICCV'23):** Designed a Curvature-Aware Geometric Black-Box Attack exploiting local boundary geometry, achieving state-of-the-art success across image classifiers.
- **Generative Transferable Attack (Under Review, ICLR'26):** Proposed a generative attack to generate confident adversarial examples leveraging pruned-ensemble discriminators + core target samples that are highly transferable to deceive any unknown victim model.
- **Federated Learning (Under Review, ICASSP'26):** Designed mobility-aware decentralized learning frameworks that mitigate catastrophic forgetting and improve training stability under network constraints.
- **LLM/VLM Jailbreaks (Ongoing):** Leading responsible red-teaming of large multimodal models to evaluate safety alignment under single corrupted prompt exposure.
- **Alignment Research (Current Focus):** Investigating LoRA/QLoRA fine-tuning and RLHF/DPO/ORPO post-training methods for LLMs, with emphasis on controllability, refusal behavior, and safety trade-offs.

SELECTED PROJECTS

Cross-Modal Jailbreak Pipeline: Implemented end-to-end diffusion sampling, latent conditioning, and prompt-image co-design to produce images that cause misalignment in VLMs; built evaluation suite measuring success rate, semantic fidelity, and transferability across multimodal models.

Terrain Identification: Developed CNN and time-series models to classify human activities using imbalanced accelerometer and gyrosopic data.

Feature Learning by Inpainting Using GAN: Designed GAN-based models to restore missing image pixels.

Eye Cataract & Glaucoma Detection: Designed a classification model using Logistic Regression, Random Forest, and SVM to identify cataract and glaucoma conditions.

Banknote Authentication: Implemented ANN and Decision Tree classifiers for fraud detection.

Data-Driven Classification Models: Implemented variety of classifiers for pattern recognition tasks, including linear and quadratic regression, logistic regression, k-nearest neighbors, naïve Bayes, and optimal Bayes.

SELECTED PUBLICATIONS

- [1] **M. F. Reza**, R. Jin, T. Wu, and H. Dai, "GSBA^K: top-K Geometric Score-based Black-box Attack." Accepted in *Proceedings of International Conference on Learning Representations (ICLR)*, 2025.
- [2] **M. F. Reza**, A. Rahmati, T. Wu, and H. Dai, "CGBA: Curvature-aware Geometric Black-box Attack." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [3] **M. F. Reza**, R. Jin, T. Wu, M. F. Pervej, and H. Dai, "Boosting Targeted Adversarial Transferability: A Generative Approach Using Pruned Ensembles and Core Target Samples." **Under review** in *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- [4] **M. F. Reza**, et al., "Distribution-Aware Mobility-Assisted Decentralized Federated Learning" **Under review** in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
- [5] **M. F. Reza**, et al., "Jailbreak Attacks that Bypass LLM Safety with Minimal Tuning" **Ongoing work**.

LEADERSHIP AND SERVICE

Participation: ICCV23, TENSYP 2020, EICT 2019, iCEEiCT 2018, ICECTE 2016.
Reviewer: CVPR2025, ICLR2025, NeurIPS 2024, PEEIACON2024, ICECTE, ICEEE.
Memberships: IEEE Graduate Student Member; Student Member of IEEE - Computer Society.

RESEARCH INTERESTS

Computer Vision, Adversarial Attacks and Defense, Deep Learning, Federated Learning, Generative AI, NLP.

EDUCATION

North Carolina State University <i>Doctor of Philosophy</i> in Electrical Engineering; GPA: 4.00/4.00 Advisor: <i>Prof. Huaiyu Dai</i>	Raleigh, NC Fall 2021 - Present
Rajshahi University of Engineering & Technology Master of Science in Electrical & Electronic Engineering; GPA: 4.00/4.00 Advisor: <i>Prof. Md Selim Hossain</i>	Rajshahi, Bangladesh May 2015 - Oct. 2017
Rajshahi University of Engineering & Technology Bachelor of Science in Electrical & Electronic Engineering; GPA: 3.90/4.00 (1st among 123 students)	Rajshahi, Bangladesh Mar. 2010 - Dec. 2014

TEACHING EXPERIENCE

Graduate Teaching Assistant , Department of ECE, NC State University, Raleigh, NC	Fall 2021 - Present
- Assisted in courses: Signals and Systems; Introduction to Signals, Circuits and Systems; Electric Circuits. - Designed lab assignments and evaluation rubrics; mentored students in machine learning projects.	
Assistant Professor , Rajshahi University of Engineering & Technology, Bangladesh	Mar., 2018-Present(on leave)
- Taught undergraduate courses: Machine Learning; Digital Signal Processing; Numerical Analysis; Electronics; Power Electronics. Supervised 6 undergraduate final-year theses (primary): guided topic selection, methodology, experiments, and thesis writing; produced 2 posters and 1 manuscript submission.	
Lecturer , Rajshahi University of Engineering & Technology, Bangladesh	Sep., 2015-Feb., 2018
Taught undergraduate courses: Linear Systems; Electrical Circuits; Electrical Machines.	

GRADUATE COURSEWORK

Taken at NC State University: Theoretical Foundation of Large-Scale Machine Learning, Pattern Recognition, Neural Networks, Optimizations and Algorithms, Introduction to Wireless Networking.

Completed at Rajshahi University of Engineering & Technology: Engineering Analysis, Information and Coding Theory, Antenna Array Signal Processing, Digital Signal Processing, Computational Electromagnetics, Antennas and Propagation.

HONORS & AWARDS

Graduate Merit Award , NC State University, Raleigh, NC	2024
Travel grants , NC State University, Raleigh, NC	2023
University Gold Medal , Rajshahi University of Engineering & Technology, Bangladesh	2019
Prime Minister Gold Medal , University Grants Commission, Bangladesh	2017
M.Sc. Scholarship , Rajshahi University of Engineering & Technology, Bangladesh	2015 - 2017
Student of the year awards for securing highest CGPA in both 2nd and 4th years of B.Sc.	2012, 2014